

HARSH JHA

Node.js Backend Developer | Express · MongoDB · Redis · PostgreSQL · AWS

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PROFESSIONAL SUMMARY

Backend developer specializing in **Node.js** and **Express**, with hands-on experience building production-grade, transaction-safe financial systems and large-scale web automation pipelines. Skilled in designing concurrency-safe data models, **RESTful APIs**, and secure authentication flows across **MongoDB, Redis, and PostgreSQL**. Delivered automation covering **100+ U.S. county systems** and a fintech ledger backend with **zero-overdraft guarantees** under simultaneous load.

EXPERIENCE

Backend & Automation Engineer Intern | *Invicto Labs*

Aug 2025 – Dec 2025 · Bangalore, India

- Built **Node.js/Puppeteer** automation pipelines to retrieve property tax and mortgage records across **100+ U.S. county government websites**, replacing repetitive manual lookups with a repeatable verification workflow.
- Designed multi-step verification workflows to cross-check tax dues and payment history against unique property identifiers, feeding verified results into internal mortgage-approval workflows.
- Built **Express.js APIs** to trigger and monitor county-specific automation jobs on demand and integrate results into internal mortgage-verification systems.
- Handled site-specific edge cases across **100+ government portals**, improving automation reliability and reducing manual re-runs.

Full Stack Developer Intern | *TRIOT Solutions*

Jan 2025 – Jul 2025 · Remote, India

- Built the **React** frontend for a ride-booking platform from scratch using **React, Tailwind CSS, React Router, and Zustand**, with reusable components and responsive layouts.
- Contributed to the **Spring Boot** backend by developing **REST APIs**, integrating **PostgreSQL** with role-based access control, and securing authentication with **BCrypt** password hashing.
- Collaborated in an Agile environment to integrate backend REST APIs with the React frontend, delivering end-to-end ride-booking workflows.

PROJECTS

Nivora Ledger — Production-Grade Fintech Backend

Node.js · Express.js · MongoDB · Redis · JWT · Zod · Jest · Docker · GitHub Actions · GHCR | [GitHub](#)

- Engineered a double-entry, append-only ledger with balances derived from immutable entries, preventing direct balance mutation and eliminating balance drift by construction.
- Built concurrency-safe, idempotent fund transfers using **MongoDB transactions** and account-level locking/versioning, preventing duplicate transfers and overdrafts under simultaneous requests.
- Implemented administrator-authorized transaction reversal with atomic compensating ledger entries while preserving original transactions for auditability.
- Secured financial operations with **JWT authentication**, HTTP-only cookies, token blacklisting, system-user authorization, **Zod** validation, rate limiting, and structured **Pino** logging.
- Built a **43-test Jest/Supertest suite** covering authentication, transfers, concurrency, idempotency, rate limiting, ledger immutability, transaction history, and reversal; containerized with **Docker** and automated CI/CD with **GitHub Actions + GHCR**.

TECHNICAL SKILLS

- Languages:** JavaScript, TypeScript, Java
- Backend:** Node.js, Express.js, Spring Boot, REST APIs, Zod, JWT
- Frontend:** React.js, Redux, Zustand, React Router, Tailwind CSS
- Databases:** MongoDB, MongoDB Atlas, PostgreSQL, MySQL, Redis
- Testing:** Jest, Supertest
- DevOps & Infrastructure:** Docker, Docker Compose, GitHub Actions, GitHub Container Registry, AWS (EC2, S3), Linux
- Concepts:** Distributed Systems, Concurrency Control, Idempotency, Double-Entry Ledger, Rate Limiting, Authentication & Authorization
- Tools:** Git, GitHub, Swagger/OpenAPI, Postman, Pino

EDUCATION

Jain University — MCA

Jul 2024 – Present, Bangalore, India

Girijananda Chowdhury University — BCA

Aug 2020 – Jul 2023, Guwahati, India